THE HENSELIAN LOCAL RING OF A DECENT SPACE AT A POINT IS THE HENSELIZATION OF AN ELEMENTARY ETALE NEIGHBOURHOOD

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Editorial note. Counted span is the author's own proof environment, source lines 2744-2759, immediately following the statement at 2731-2742.

Editorial note. One cross-chapter pointer pair (More on Morphisms, Categories) and one same-chapter stub are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Elementary etale neighbourhoods form a cofiltered category. Lemma lemma-elementary-etale-neighbourhoods of this chapter, cited at proof lines 2745-2746. Cofilteredness is the structural input: it makes the neighbourhood category of (U,u) initial inside that of (X,x).
- Henselization as a colimit over elementary etale neighbourhoods. More on Morphisms, Lemma more-morphisms-lemma-describe-henselization, cited at proof lines 2749-2750. Identifies the henselian local ring with the colimit over the opposite of the neighbourhood category – the bridge that turns 'initial' into 'cofinal'.

Source reference. Original reference 04E7: focused-seeds/geometry/S9-decent-describe-henselian-local-ring/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BGV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2677–2677. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05KS: focused-seeds/geometry/S9-decent-describe-henselian-local-ring/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $x \in |X|$. Let $(U, u) \to (X, x)$ be an elementary étale neighbourhood. Then*

$$\mathcal{O}^h_{X,x} = \mathcal{O}^h_{U,u}$$

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In words: the henselian local ring of X at x is equal to the henselization $\mathcal{O}_{U,u}^h$ of the local ring $\mathcal{O}_{U,u}$ of U at u.

Proof. Since the category of elementary étale neighbourhood of (X, x) is cofiltered (Lemma 0BGV) we see that the category of elementary étale neighbourhoods of (U, u) is initial in the category of elementary étale neighbourhood of (X, x). Then the equality follows from More on Morphisms, Lemma 05KS and Categories, Lemma 04E7 (initial is turned into cofinal because the colimit defining henselian local rings is over the opposite of the category of elementary étale neighbourhoods). □